

Final Taxonomy:

Coding Principle:

Each human-authored commit is assigned exactly one primary category based on its dominant purpose. Evidence should be weighed in this order: (1) patch content and behavioral effect, (2) filenames and file roles, (3) the commit message text itself. The PR's merged/closed outcome is not considered.

This taxonomy consolidates eleven mutually exclusive categories, each defined below with inclusion/exclusion rules, message and patch indicators, one representative example with PR ID and commit SHA.

- **Functional Correctness Repair:**
 - **Definition:** The human corrects existing implementation behavior that is incomplete, incorrect, inconsistent with intended semantics, or mishandles a runtime case. The capability already exists; the human changes how it works so it behaves correctly.
 - **Inclusion criteria:** Corrects an incorrect condition, state transition, or control-flow decision; handles a previously omitted or mishandled runtime case; repairs incorrect data relationships, parameters, or API behavior; restores intended semantics of an existing feature without primarily adding new capability.
 - **Exclusion criteria:** Excludes cases where the main problem is compilation, packaging, or platform compatibility, where the change is primarily visual/interactive, where only tests are corrected, or where the change primarily introduces new capability.
 - **Common indicators:** Messages containing fix, correctly handle, handle ..., missing, fix relations, fix ... parameters.
 - **Representative examples:**
 - i. PR 2788912798, SHA e71e6e0baf2453f309688e46646d07299bd5d8baL "fix: Update ButtonGroupWidget to correctly handle button disabling based on form validity"; the human corrects the validity condition and propagates required form-validity state.
 - ii. PR 2816690510 SHA 4ccecb8b04ffd0a1adc3004ff68fbe0c27967420 "Also handle rows w/o sections." The patch adds handling for Airtable records that produce no sections instead of incorrectly constructing a document.
 - iii. PR 2896059278 SHA 7d797f55b2a7e90e2c226c1a6200fc5bb04a9fe0 "fix relations." The patch corrects database relation definitions and foreign-key relationships.
- **Feature Completion or Extension**
 - **Definition:** The human adds meaningful functionality, completes an incomplete agent-generated feature, or extends the contribution with a new capability, operation, or supported use case.

- **Inclusion criteria:** Introduces a new operation, option, endpoint, workflow, or capability; adds support for an additional use case; completes substantive functionality absent from the agent's contribution; exposes new behavior to callers or users.
- **Exclusion criteria:** Excludes correction of already-intended behavior, purely visual or interactivity polish of existing capabilities, configuration changes needed only to make existing code build
- **Common indicators:** feat, add, support, enable, implement, introduce, allow; new public methods, options, handlers, workflows, or substantial new implementation blocks.
- **Representative examples:**
 - i. PR 2761376844, SHA 0c6f9a7cff50bcae00cef19f8e515fc7f3f67d7e, "feat: add optional return_html flag to SeleniumScrapingTool"; adds a new controllable option.
 - ii. PR 2766442004 - SHA 1673e5b7437b7fac2ca734c6e4bcd077d8e34564 - "feat(desktop/agents): goal planner." Adds and develops a goal-planning agent capability.
 - iii. PR 2821578095 - SHA c3d11c3b25d49a4521b1b96eecf754ae833ef96b - "feat: add close()." Adds a new client method for explicitly closing the WebSocket connection and demonstrates its use.
- **Test and Validation Adjustment**
 - **Definition:** The human's dominant intervention is adding, repairing, strengthening, or adapting tests or other validation artifacts used to verify the agent-generated contribution.
 - **Inclusion criteria:** New unit, integration, regression, or end-to-end tests; corrected assertions or expected outputs; test cases adapted to intended behavior; fixture or test-configuration changes made primarily to improve validation.
 - **Exclusion criteria:** Excludes cases where production behavior is substantially corrected and tests merely accompany that fix, excludes formatting-only test changes, excludes deletion of obsolete tests where deletion is the dominant purpose.
 - **Common indicators:** test, tests, unit test, Cypress, pytest, e2e, spec; additions of assertions, fixtures, or mock data concentrated in test directories.
 - **Representative examples:**
 - i. PR 2976106239, SHA c7c886e149ce93f486622043f7937a02b0b26896, "add Patch team routing form responses e2e"; adds multiple end-to-end cases including invalid inputs and update behavior.
 - ii. PR 2791971139 - SHA 3e5bab148ad610b2d40ad8541f8268583e64e100 - "fix: Update cypress test cases." Changes Cypress expectations and interactions to match the intended autocomplete behavior.
 - iii. PR 2905670458 - SHA e1acf93987ef21b0671c18c66aa266307b5e4be0 - "test also OpArg.static_type." Extends a compiler test to verify an additional property.

- Structural Refactoring
 - **Definition:** The human reorganizes, renames, extracts, simplifies, or redistributes implementation structure while substantially preserving intended external functionality.
 - **Inclusion criteria:** Moves or renames implementation components; extracts classes/modules/functions; reorganizes imports and module boundaries; restructures architecture or simplifies internal organization without intentionally eliminating functionality.
 - **Exclusion criteria:** Excludes primarily deleting unnecessary material, deliberate runtime-semantics changes, formatting-only cleanup
 - **Common indicators:** refactor, rename, move, extract, reorganize, simplify; import updates, file relocation, module splitting, or responsibility redistribution.
 - **Representative examples:**
 - i. PR 2758742372, SHA 30ef93f0ce59cf28e91c9f2d14c078e089a122cb; "rename + refactor"; moves components and systematically updates import paths across many files.
 - ii. PR 2894930869 - SHA 709941c4c77ea87e106af84f21c28ba6410cf06f - "Refactor LLM module by extracting BaseLLM to a separate file." Extracts the base abstraction into a dedicated module and rewires imports.
 - iii. PR 2895262047 - SHA 821a1eec7184e8a487cfcb4522d78706e5aac7bf - "refactor: Reorganize policy comment action and form import paths." Reorganizes source locations/import relationships rather than changing the feature's purpose.
- Code Cleanup
 - **Definition:** The human removes unnecessary, obsolete, duplicate, generated, or over-engineered portions of the contribution so the resulting change is smaller or simpler
 - **Inclusion criteria:** Removes unused parameters, functions, imports, files, or configuration; deletes unnecessary calls, abstractions, or debug/logging statements; removes obsolete or deprecated material and accidental artifacts
 - **Exclusion criteria:** Excludes functionality moved elsewhere, formatting/lint-only cleanup, explicit reversal of a prior commit, removal that repairs incorrect runtime behavior
 - **Common indicators:** remove, unused, unnecessary, redundant, delete, cleanup; patches dominated by deletions, removed debug statements, or deleted duplicate definitions.
 - **Representative examples:**
 - i. PR 2916133951, SHA df84e910285179eef2dc4a280701451a25ec72b7, "Remove unused localization files and clean up translation code"; deletes unused generated resources.
 - ii. PR 2795172513 - SHA ea178fb6acb414047e7574542009412db76bef58 - "Removed unused API key parameter." Eliminates an unnecessary configuration parameter throughout the SPL token plugin path.

- iii. PR 2816855074 - SHA da28bc63c10762e93c4f3309c96eb39f017522fd - "Remove unnecessary call." Removes an unnecessary early channel lookup and simplifies the surrounding logic.
- User-Facing Interface and Interaction Refinement
 - Definition: The human changes how existing functionality is displayed, styled, exposed, or interacted with, without primarily introducing new underlying domain capability.
 - **Inclusion criteria:** Styling and layout adjustments; visibility changes; interaction behavior such as menus, dialogs, clicks, hover, or tooltip presentation; usability improvements; user-facing text or localization changes.
 - **Exclusion criteria:** Excludes incorrect application-state or business logic, entirely new user capabilities, source-code formatting unrelated to product appearance
 - **Common indicators:** styling, layout, design, UI, display, hide, show, menu, modal, popover, tooltip, interactivity; CSS/class changes, visibility-state toggles, and localized user-facing strings.
 - Representative examples:
 - i. PR 2769559533 - SHA 6b138a20721ca089e0dd43ab756cbd3be937e161 - "Hide image button"; changes the visibility of an existing chat-input control.
 - ii. PR 2840501983 - SHA 3650b9e6d1e5d75ee2553046179c86bec926514e - "improved styling of menu and right-click functionality." Adjusts cursor/interaction behavior and right-click-menu implementation.
 - iii. PR 2855360043 - SHA ade0a777b6b89d53b33dd7b1db26f01f7f3e4a2e - "better layout and interactivity." Restructures settings-modal presentation and interactive controls.
- Project Integration and Environment Maintenance:
 - **Definition:** The human modifies the technical scaffolding required for the contribution to build, run, package, test in CI, or fit the project's dependency and platform ecosystem, encompassing build, CI, environment, dependency, and version maintenance.
 - **Inclusion criteria:** Build-system corrections; CI/workflow changes; platform-specific incompatibilities and macros; incorrect environment or resource paths; startup/package configuration failures; dependency additions, removals, or version bumps; lockfile/manifest refreshes.
 - **Exclusion criteria:** Excludes ordinary application/domain runtime bugs, new product behavior involving configuration, generated files regenerated from canonical sources, formatting fixes within configuration files
 - **Common indicators:** build, CI, workflow, config, dependency, package, version, bump, docker, .env, CMake; workflow files, package manifests, lockfiles, build flags, and platform attributes.
 - **Representative examples:**

- i. PR 2884694604 - SHA 0fd8b2033bf965025ede64d3a9670075b52dff45 - "Fix FATFS compiling"; adds preprocessor handling so code compiles under that configuration.
 - ii. PR 2991480772 - SHA 3ba8a9d49a2dbe03bb547792d5be04701f4948de - "Fix knapsack in pull_request_target CI" Adds the correct PR head GITHUB_REF and GITHUB_SHA values to the CI job.
 - iii. PR 2990679795 - SHA 365e44e6383173dfd134bc119a1131cb2487edf3 - "Update package.json" Updates the package version from 1.22.1 to 1.23.0, a packaging/version-maintenance intervention.
- Code Convention Conformance:
 - **Definition:** The code is necessary and behaviorally acceptable, but the human modifies its non-functional form to conform to project-level coding conventions or mechanical quality requirements like formatting, linting, static-analysis, or source-style expectations
 - **Inclusion criteria:** Formatter output; lint-rule corrections; whitespace normalization; import ordering; mechanically equivalent syntax preferred by a linter; static-type corrections that do not intentionally modify runtime semantics.
 - **Exclusion criteria:** Excludes substantive code redesign, removal of genuinely unused functionality, type changes that fix actual runtime defects, UI styling changes
 - **Common indicators:** lint, format, mypy, clippy, prettier, eslint, style, whitespace, type issue, import rearrangement, and line wrapping.
 - **Representative examples:**
 - i. PR 2788912798, SHA e2b048560930589c1d3695d869136ed4a19f953e, "Fixes linting issues"; primarily reformats expressions and removes trailing whitespace.
 - ii. PR 2958489417 - SHA d0fc4fec5799cab14e39e9af95a1664ffd07b938 - "format" Reorders imports/exports without changing behavior.
 - iii. PR 2979615970 - SHA da65fa5944e7b0da010f1dcbc7b600d88f74f429 - "lint fix" Removes trailing whitespace, adds preferred callback parentheses, and wraps a long expression.
- Documentation:
 - **Definition:** The human updates explanatory, instructional, developer-facing, or release-facing information surrounding the contribution rather than its runtime behavior.
 - **Inclusion criteria:** README and documentation updates; developer comments; changelog and changeset/release-note additions; explanatory source comments where documentation is the dominant purpose; usage examples intended primarily to explain functionality.
 - **Exclusion criteria:** Excludes user-visible application text or translations; package version changes themselves, executable examples whose main purpose is verification
 - **Common indicators:** docs, README, changeset, changelog, comment, documentation; Markdown files, release-note files, and explanatory comments.

- **Representative examples:**
 - i. PR 2761376844, SHA 20cdb5690f3e0a7e000583cd0fc15ee58ab1fb45, "chore: update readme"; updates documentation describing tool behavior and options.
 - ii. PR 3003168258 - SHA 32dd7876dba812cc1ccc2580fe4ded9eccb31bda - "update docs" Updates DASHBOARD.md to clarify routing, data organization, and application behavior.
 - iii. PR 2947870333 - SHA 7dc5e4dfbe2086875bf75bf0b9e868267bbd8ce - "Add changeset" Adds patch-release metadata communicating that the package should be released.
- **Version-Control Conflict Resolution**
 - **Definition:** The human reconciles the agent-generated contribution with diverging repository history by resolving merge or rebase conflicts rather than independently implementing new work.
 - **Inclusion criteria:** Commit explicitly states conflicts are being resolved; patch spans divergent changes from multiple branches; the dominant purpose is reconciliation rather than implementation.
 - **Exclusion criteria:** Excludes normal functional fixes made after a merge, explicit reversal of a prior commit, ordinary refactoring that happens to touch many files
 - **Common indicators:** resolve conflict, resolve conflicts, merge conflict, conflict resolution; unusually broad cross-file synchronization mixing competing branch states.
 - **Representative examples:**
 - i. PR 2999563616, SHA f391335b7b78dfa1b5f2c33f9d779ff85adc494f , "resolve conflicts"; reconciles eight schema/editor and dependency files.
 - ii. PR 2999563616 - SHA a2f0fb8a25b31ca942ec0126d5f7e41048484249 - "resolve conflict" Reconciles changes across 69 files, including project features, configuration, package metadata, and documentation.
 - iii. PR 3013906937 - SHA 7b6468f58757b676c76e04b9c5d5d64f092a70f0 - "resolve conflicts" Reconciles a very broad set of changes across 142 files.
- **Rollback or Reversion**
 - **Definition:** The human intentionally restores a previous state by explicitly undoing an earlier commit or change, regardless of what type of work the reverted commit originally performed.
 - **Inclusion criteria:** Explicit Git revert commits; commits clearly described as reverting or rolling back an earlier change; deliberate restoration of the immediately preceding implementation.
 - **Exclusion criteria:** Excludes ordinary deletion of unnecessary code, replacement of a flawed implementation with a corrected one, resolution of competing branch versions
 - **Common indicators:** "Revert ...", revert, rollback, back out, "This reverts commit ..."; patches that mechanically invert an earlier change or restore recently deleted code.

- **Representative examples:**

- i. PR 2821716083 - SHA 30bc938a929f0f8111e287b847a246eab4f6119d - Revert "fix: regenerate poetry.lock for Python 3.11 compatibility"; restores prior dependency versions.
- ii. PR 2972319075 - SHA 7eb353f92fec1b6fe16f053c2cc13037230d065b - Revert "Remove preview sections from assigneesPage.tsx" Explicitly restores UI/assignment behavior removed by the referenced earlier commit.
- iii. PR 2985471045 - SHA 0d5385611582bf72ea28f6c6a9e1896963fe6e14 - Revert "fix: add service_role bypass policy for tests" Explicitly removes the service-role bypass policy introduced by an earlier change.
- iv. PR 3010545410 - SHA c703d3ca315e2a9703730822e5edfcefde37cdc5 - Revert "Update Supabase generated files" Explicitly restores the earlier state of the generated database schema/type artifacts.